1 Additional Experiments on Orthogonal Least Squares Regression (OLSR)

Orthogonal Least Squares Regression (OLSR) [1] is a supervised manifold-constrained learning problem widely used for feature extraction in linear discriminant analysis.

Consider a distributed setting with n agents, where each agent has access to m samples with d features. The distributed OLSR problem is formulated as

$$\min_{X \in \mathbb{R}^{d \times r}} \frac{1}{n} \sum_{i=1}^{n} \|P_i X - Q_i\|_F^2 \quad \text{s.t. } X \in \mathrm{St}(d, r),$$

where $P_i \in \mathbb{R}^{m \times d}$ and $Q_i \in \mathbb{R}^{m \times r}$ denote the local data matrices at agent i.

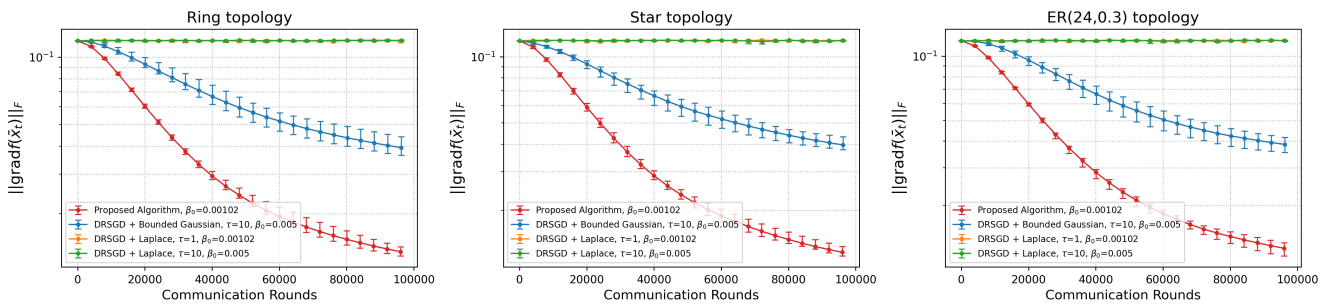

Figure 1: Experimental results on synthetic data for OLSR, measured by the Frobenius norm of the evaluated Riemannian gradient. The number of agents is $n = 24$, the number of communication rounds is denoted by τ as indicated in the legend, and the stepsize is selected as $\beta_t = \beta_0/\sqrt[3]{10^{-4}t + 1}$. The three panels, from left to right, correspond to the ring, star, and Erdős–Rényi (ER) communication graphs, respectively.

2 Additional Experiments Comparing with the Standard Gaussian Mechanism

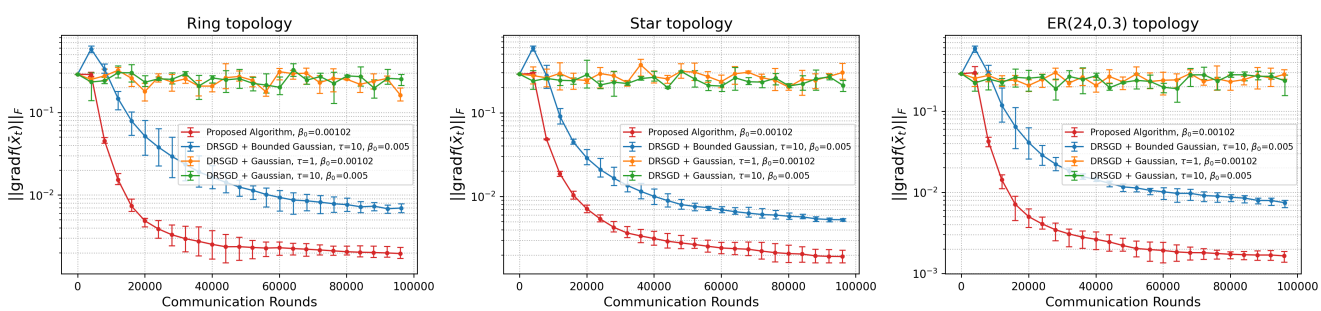

Figure 2: Experimental results on synthetic data for PCA, measured by the Frobenius norm of the evaluated Riemannian gradient. The number of agents is $n = 24$, the number of communication rounds is denoted by τ as indicated in the legend, and the stepsize is selected as $\beta_t = \beta_0/\sqrt[3]{10^{-4}t + 1}$. The three panels, from left to right, correspond to the ring, star, and Erdős–Rényi (ER) communication graphs, respectively.

3 Additional Experiments on CPU Time and Memory Usage

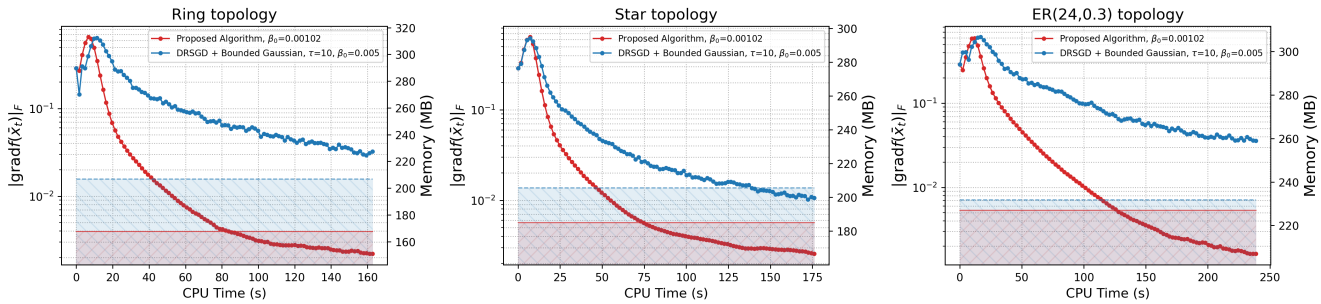

Figure 3: Experimental results on synthetic data for PCA from a single run. The left y-axis shows the Frobenius norm of the evaluated Riemannian gradient (curves), the right y-axis shows memory usage (shaded regions), and the x-axis represents CPU time. The number of agents is $n = 24$, the number of communication rounds is denoted by τ as indicated in the legend, and the stepsize is selected as $\beta_t = \beta_0/\sqrt[3]{10^{-4}t+1}$. The three panels, from left to right, correspond to the ring, star, and Erdős–Rényi (ER) communication graphs, respectively.

4 Additional Experiments on the Robustness of the Algorithm

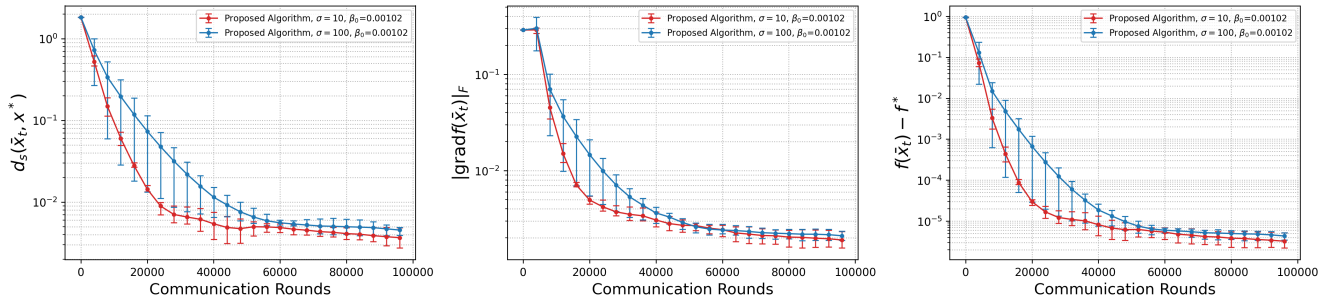

Figure 4: Experimental results on synthetic data for PCA over the ring communication graph. The three panels show, from left to right, the subspace distance between $\bar{x}_t$ and the optimal solution, the Frobenius norm of the Riemannian gradient, and the objective gap to the optimal value $f(\bar{x}_t) - f^*$. The number of agents is $n = 24$, and the stepsize is chosen as $\beta_t = \beta_0/\sqrt[3]{10^{-4}t+1}$. We compare two variance settings, $\sigma = 10$ and $\sigma = 100$, under the same $\beta_0 = 0.00102$.

References

[1] Haifeng Zhao, Zheng Wang, and Feiping Nie. Orthogonal least squares regression for feature extraction. *Neurocomputing*, 216:200–207, 2016.